

Lab 05:

HW 5

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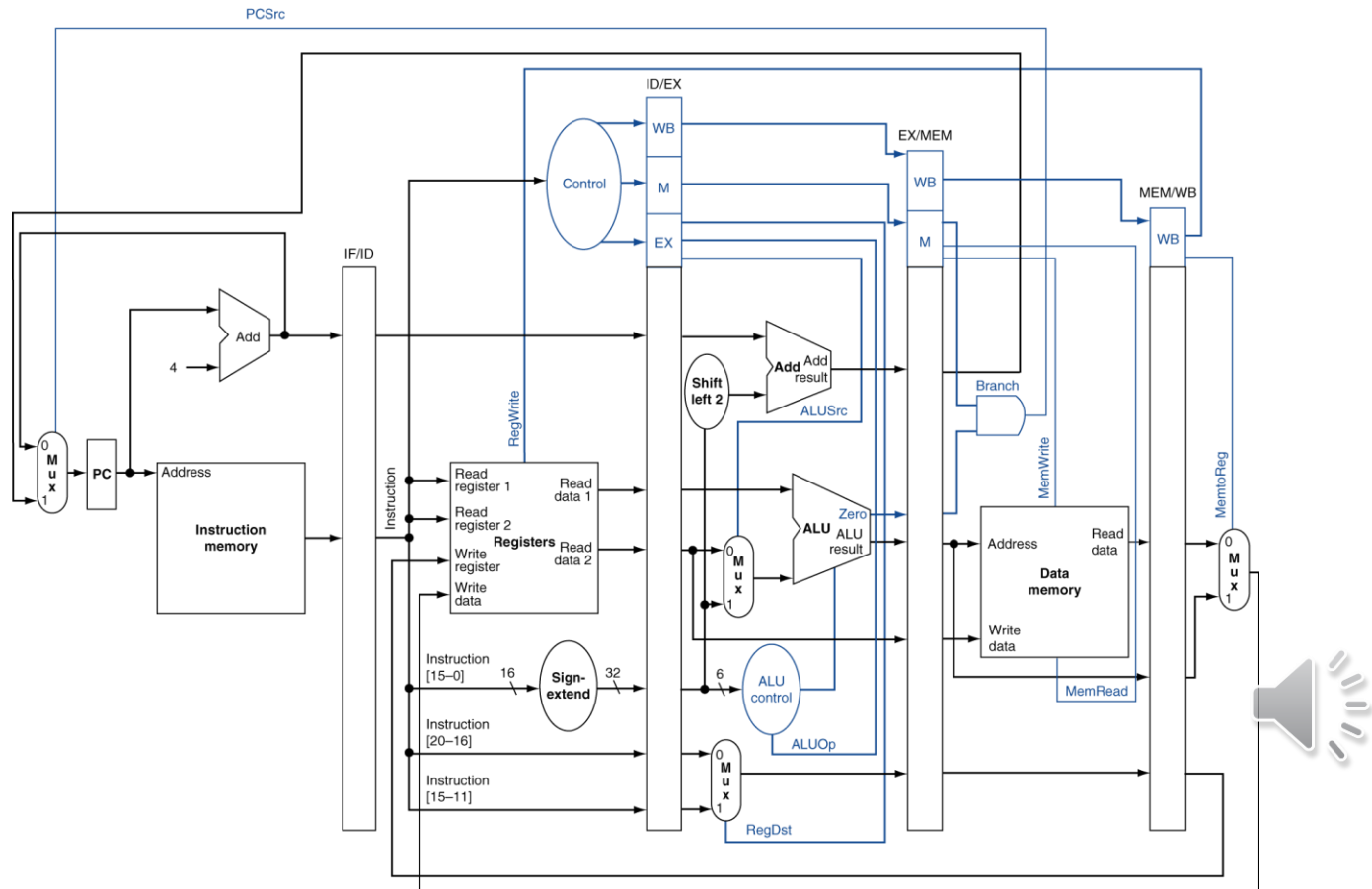
Overview

- ◆ You will upgrade your single cycle CPU into a pipelined CPU
- ◆ What to do?
 - Implement a five-stage pipelined CPU
 - We need a basic pipeline that does not support branch prediction and data forwarding
 - Predict not-taken and flush on misprediction
 - Stall upon detecting a data hazard



Target CPU Microarchitecture

- ◆ Implement a basic pipelined architecture following the CPU architecture you learned in the class
 - Add hazard detection module ...



What to implement

- ◆ #1) Divide the computation into five stages and execute multiple instructions in parallel
- ◆ #2) Properly set the control signals over the five pipeline stages (whichever way you want)
- ◆ #3) Predict not-taken during instruction fetch
 - If the branch is taken, flush the mis-predicted instructions
 - You are allowed to perform early branch resolution @ ID stage if you want
- ◆ #4) Detect data hazard and stall
 - If the CPU detects a data hazard, stall the pipeline and re-execute the ID stage (do not support data forwarding for now)



Assignment

◆ Files:

- MEM.v (HW3)
 - Use the MEM files from “HW3” again (You cannot share resources between IF and MEM stage)
- Modify other files from HW4 if you need
 - Use the files from previous HWs and modify them if you need
- Do not add your own HW module and files if you want
 - Use the ~~skeleton~~ HAZARD.v file for hazard detection and stall



Submission

- ◆ Submission (Zip all the files)
 - For a two-people team: lab5_student_id1_student_id2.zip
 - For a one-person team: lab5_student_id.zip
 - You must follow the format (-10% for wrong file format)
 - Example:
 - lab5_2020102030.zip
 - lab5_2020102030_2022103040.zip
 - The zip file should contain:
 - every v files
 - lab5_report.pdf



Submission

- ◆ You need to write a 4+-page report
 - Explain the overall structure and how you implemented each program
 - Draw the hardware modules if needed ...
 - How did you implement the hazard detection unit, and stall logic ...

- ◆ Due: Mon. May 25th
 - 1 week delay: -20%
 - Further delay: 0 point

